SAM CPU replacement board, draft 0.2, drawn from docs/ARCHITECTURE.md. Licence: CERN-OHL-S v2.
A Raspberry Pi 4 on the 40-pin header J21 runs PinMAME + ppuc (HDMI stays on the Pi). The RP2354B drives the original IO power driver board on J9 and scans the switches. Connector names and pinouts follow the original CPU/Sound board 520-5246-00. Each connector sits on the same page as the circuit that drives or reads it.
Signals between pages use global labels; the numbers in brackets next to a label list the other pages it goes to.

Power

Power: +5 V mux, external +12 V, 3.3 V, 4.5 V, monitors

File: power.kicad_sch

MCU and Pi

RP2354B SAM IO controller and Raspberry Pi 4 header

File: mcu.kicad_sch

IO bus

J9: bus to the IO power driver board (IO board J1)

File: io_bus.kicad_sch

Switch columns

Switch matrix columns: 74HC595, strobe drivers, J1

File: sw_columns.kicad_sch

Switch rows 1-8

Switch matrix returns 1-8: J6, LM339 front end, 74HC165

File: sw_rows_1.kicad_sch

Switch rows 9-16

Switch matrix returns 9-16: J12, LM339 front end, 74HC165

File: sw_rows_9.kicad_sch

Dedicated switches 1-16

Dedicated switches D1-D16 (J2, J3) and coin door memory protect

File: sw_dedicated_1.kicad_sch

Dedicated switches 17-24

Dedicated switches D17-D24 (J13) and DIP switches

File: sw_dedicated_2.kicad_sch

GI dimmer

J18 GI dimmer header

File: display_gi.kicad_sch

Audio

Audio: PCM5102A, TDA7297 stereo + subwoofer (+12 V)

File: audio.kicad_sch

Switch chain extensions

Switch-chain extensions: 7th 74HC165, first 74HC595

File: dmd_panels.kicad_sch

RS485 and I2S

Isolated RS485 port (2x RJ45), I2S input header, analog line-in

File: rs485_i2s.kicad_sch

Stern SAM CPU PPUC (CERN-OHL-S v2)

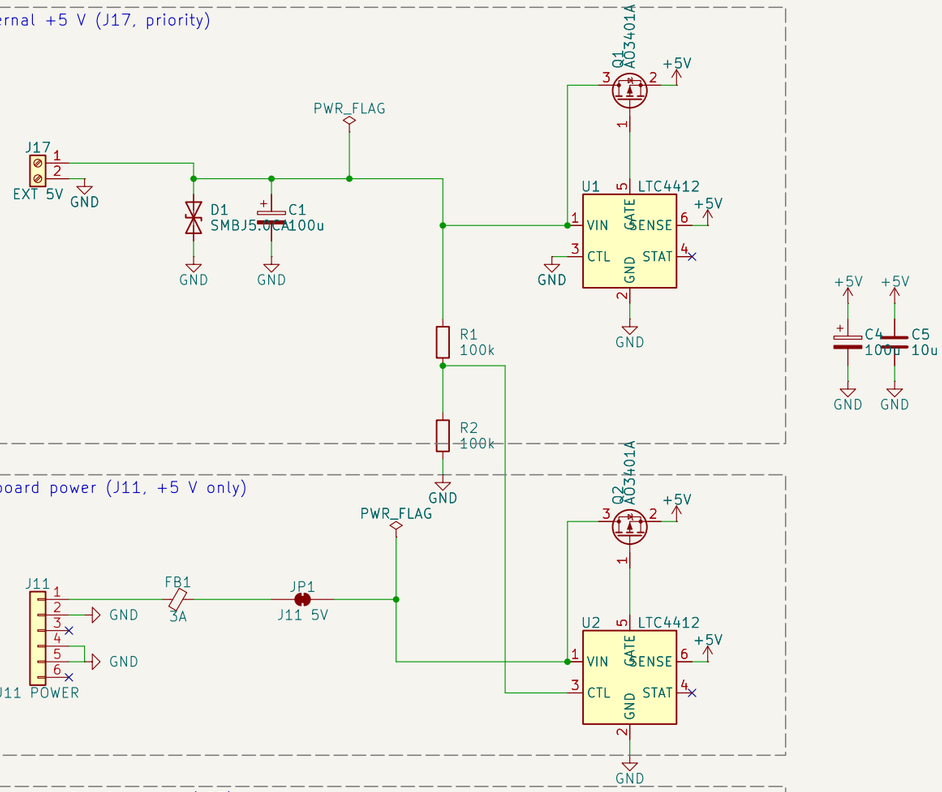
Sheet: /
File: sam_cpu.kicad_sch

Title: SAM CPU replacement board

Size: A3	Date: 2026-10-06	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 1/13

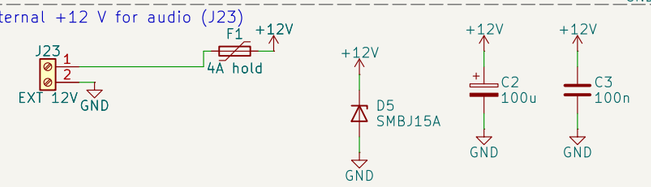
J11 keeps the original CPU board connector (IO board J16 harness) for +5 V and ground only: its +−12 V pins are not used.
The audio amplifiers run from an external +12 V supply on J23 (about 3 A at full volume). F1 + D5 blow the fuse on reverse polarity.
U3 NCP1117 (as in the RP2350 hardware guide) is stable with ceramic capacitors. All regulator capacitors are 0805 or larger for low DC-bias derating.
External +5 V on J17 has priority: U2 turns the J11 path off whenever +5V_EXT is present (EXT_PRESENT high).
LTC4412 + AO3401A ideal diodes stand in for the TPS2121 of ARCHITECTURE.md 9.3 (library part, about 4 A per path).
JP1 (bridged by default) disconnects the J11 +5 V completely; leaving J17 unplugged forces J11.

External +5 V (J17, priority)

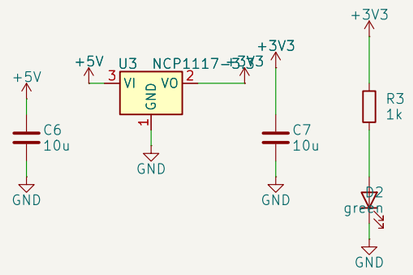


IO board power (J11, +5 V only)

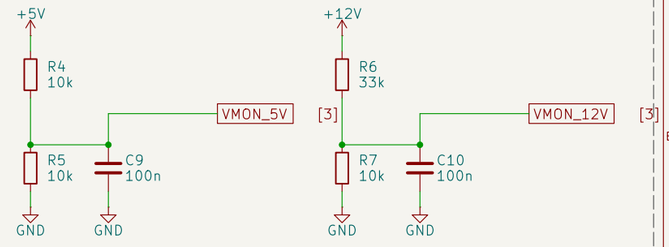
External +12 V for audio (J23)



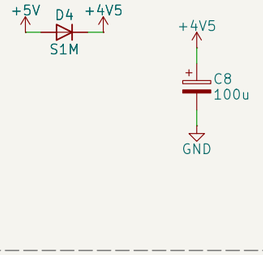
3.3 V



Supply monitors (RP2354B ADC)

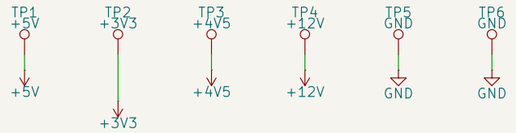


4.5 V for the switch matrix





Test points



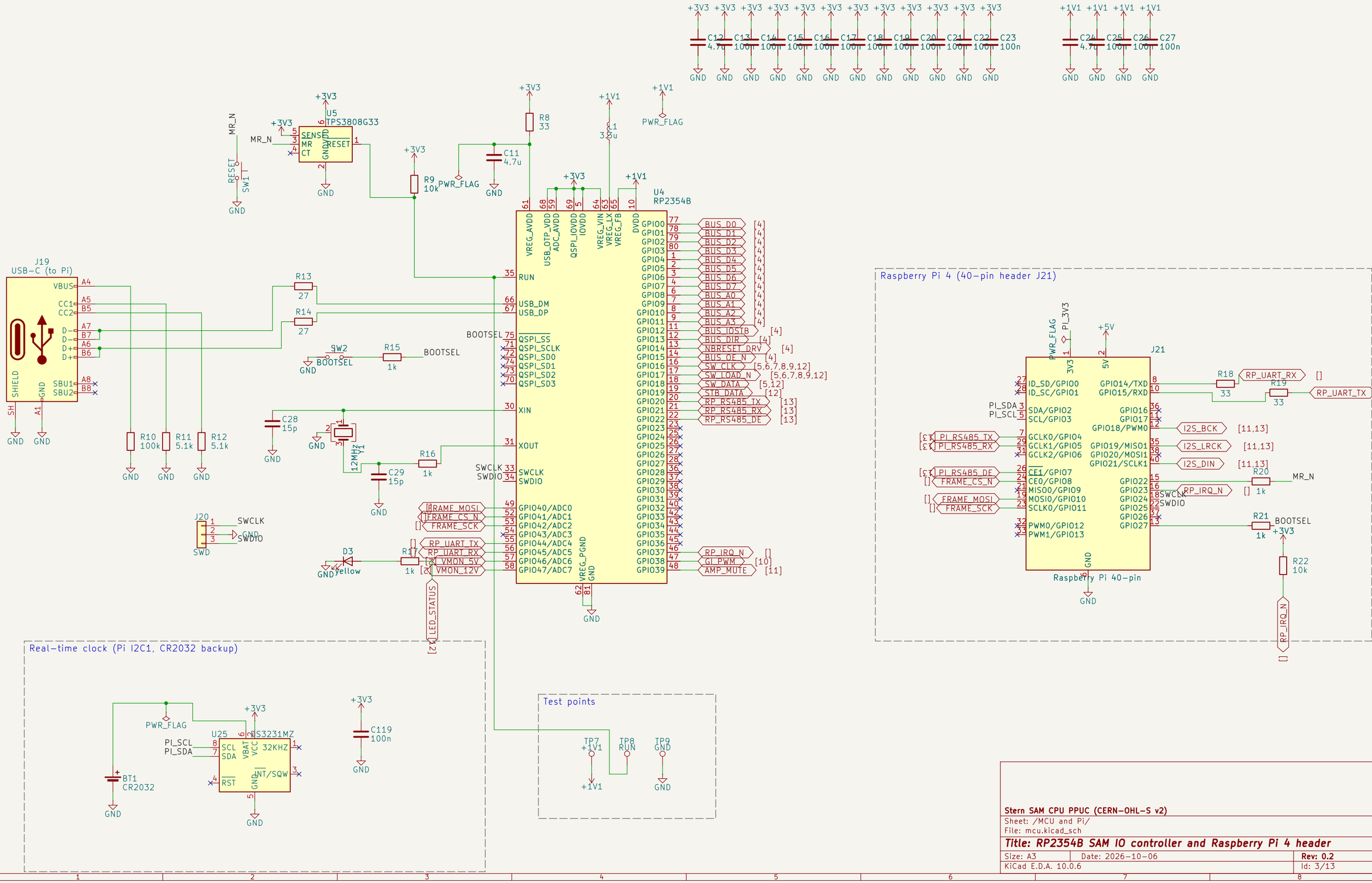
Stern SAM CPU PPUC (CERN-OHL-S v2)

Sheet: /Power/
File: power.kicad_sch

Title: Power: +5 V mux, external +12 V, 3.3 V, 4.5 V, monitors

Size: A3	Date: 2026-10-06	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 2/13

GPIO map: ARCHITECTURE.md 9.2. PIO0 bus on GPIO0–15, PIO1 switch chain on GPIO16–20, GPIO20–22: RS485 port (UART1 TX / RX, driver enable, page 13); GPIO23–36 unassigned. Pi UART3 (GPIO4 / 5, RTS on GPIO7) reaches the same port.
Frames from the Pi on SPI1 (GPIO40–42), UART0 on GPIO44–45, ADC monitors on GPIO46–47, GPIO43 spare. Coin door, status LED and GI_SPARE moved onto the switch chain.
Core supply from the internal switching regulator: VREG_LX -> L1 -> DVDD (1.1 V); VREG_AVDD through 33 R / 4.7 uF. Check against the RP2350 hardware guide.
Raspberry Pi 4 on the 40-pin header J21 (HDMI stays on the Pi). The Pi is powered from the board +5 V: never plug the Pi USB-C supply at the same time.
U25 DS3231MZ real-time clock on the Pi I2C1 bus (Linux: dtoverlay=i2c-rtc,ds3231), backed by a CR2032 in BT1.



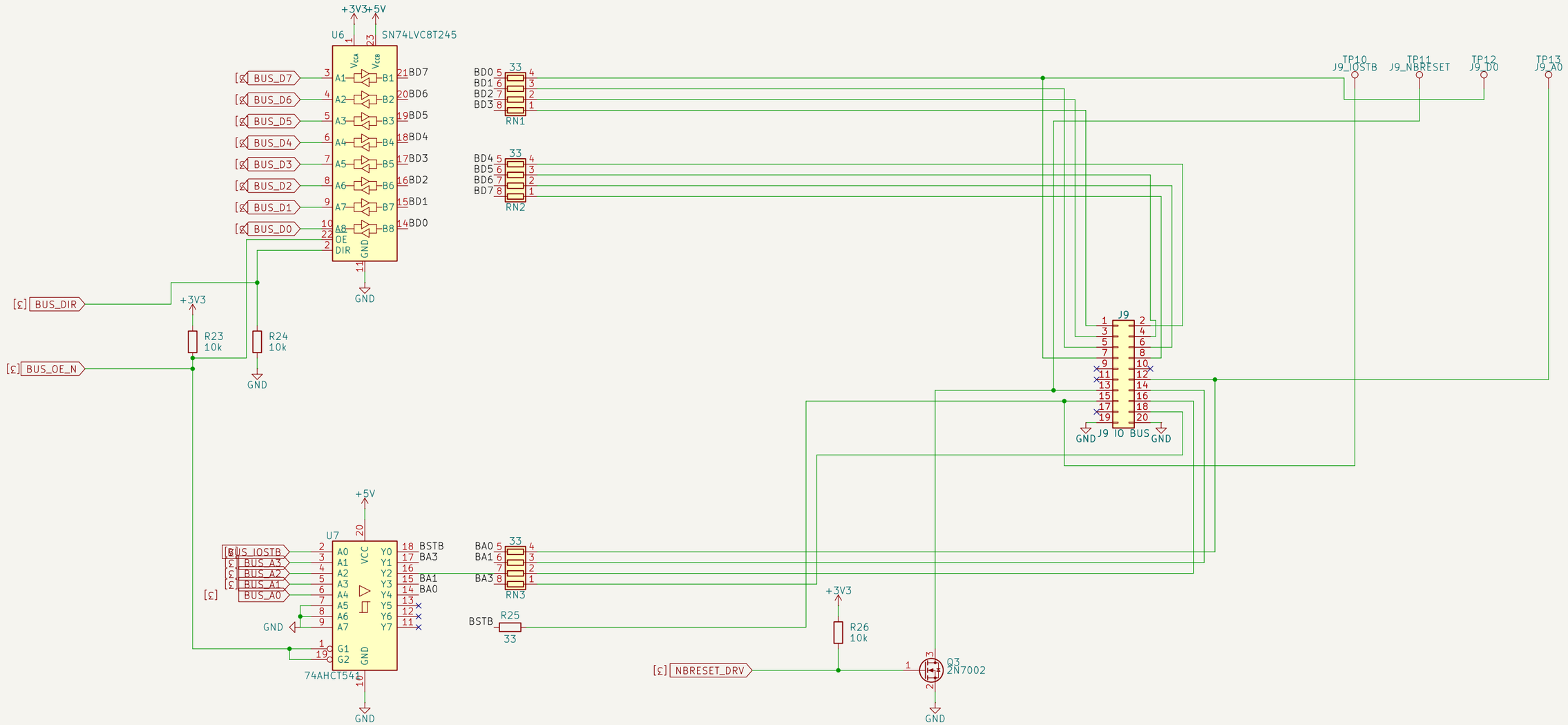
Stern SAM CPU PPUC (CERN-OHL-S v2)

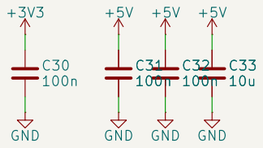
Sheet: /MCU and Pi/
File: mcu.kicad_sch

Title: RP2354B SAM IO controller and Raspberry Pi 4 header

Size: A3	Date: 2026-10-06	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 3/13

ARCHITECTURE.md 3.1–3.2. The doc calls this connector J1 (its name on the IO board); on the CPU board it is J9.
Both buffers are off at power-up (BUS_OE_N pulled high) and NBRESET is held low until the firmware releases it.
33 R series resistors on the J9 side for the ribbon cable. Unused 74AHCT541 inputs tied low.





Stern SAM CPU PPUC (CERN-OHL-S v2)		
Sheet: /IO bus/		
File: io_bus.kicad_sch		
Title: J9: bus to the IO power driver board (IO board J1)		
Size: A3	Date: 2026-10-06	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 4/13

U20 latch shares SW_LOAD_N with the 74HC165 load (one pulse: 165 loads while low, 595 latches on the rising edge). U20 now follows U31 (DMD panels sheet), which carries the status LED and GI_SPARE.
One shift clock for both registers: PIO1 shifts the next strobe pattern into the 595 while the 74HC165 chain shifts in (ARCHITECTURE.md 5.3–5.4).
Strobes: 595 output high –> MMBT3904 on –> strobe pulled low (active), with the original 1 k pull-up to +4.5 V and 100 nF.
J1 keeps the original pinout (key on pin 2). Strobes 5–8 are driven even where a game does not use them (Q15).

+4V5
R28 1k
C35 100n
R27 1k
Q11 MMBT3904
R30 1k
C36 100n
R29 1k
Q12 MMBT3904
R32 1k
C37 100n
R31 1k
Q13 MMBT3904
R34 1k
C38 100n
R33 1k
Q14 MMBT3904
R36 1k
C39 100n
R35 1k
Q15 MMBT3904
R38 1k
C40 100n
R37 1k
Q16 MMBT3904
R40 1k
C41 100n
R39 1k
Q17 MMBT3904
R42 1k
C42 100n
R41 1k
Q18 MMBT3904
GND

TP14 STB1
TP15 STB2
TP16 SW_CLK
TP17 SW_DATA
[21] SW_DATA

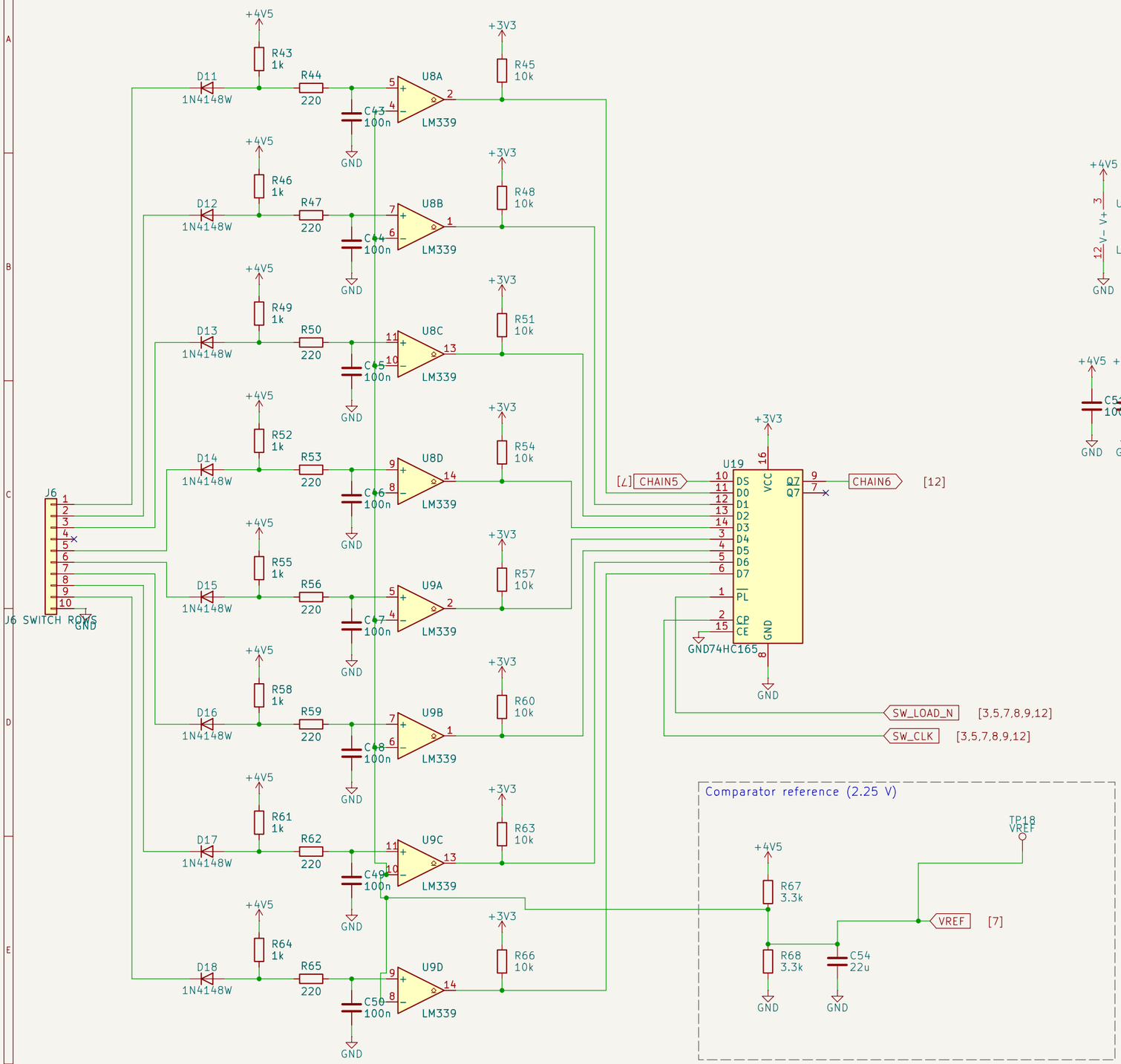
+3V3
U20 74HC595
SER
SRCLK
SRCLR
RCLK
OE
VCC
GND
QA QB QC QD QE QF QG QH QH'
[21] STB_CHAIN
[21] SW_CLK
[21] SW_LOAD_N

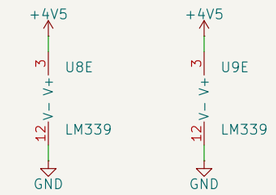
J1
J1 SWITCH COLUMNS

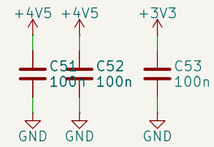
+3V3
C34 100n
GND

Stern SAM CPU PPUC (CERN-OHL-S v2)
Sheet: /Switch columns/
File: sw_columns.kicad_sch
Title: Switch matrix columns: 74HC595, strobe drivers, J1
Size: A3 | Date: 2026-10-06 | Rev: 0.2
KiCad E.D.A. 10.0.6 | Id: 5/13

Per return (ARCHITECTURE.md 5.2–5.3): series diode from the connector, 1 k pull-up to +4.5 V, 220 R + 100 nF, LM339 against VREF (2.25 V). Output 1 = open, 0 = closed, as the ROM expects. Connector pin order follows the original drawing: check it with a meter.







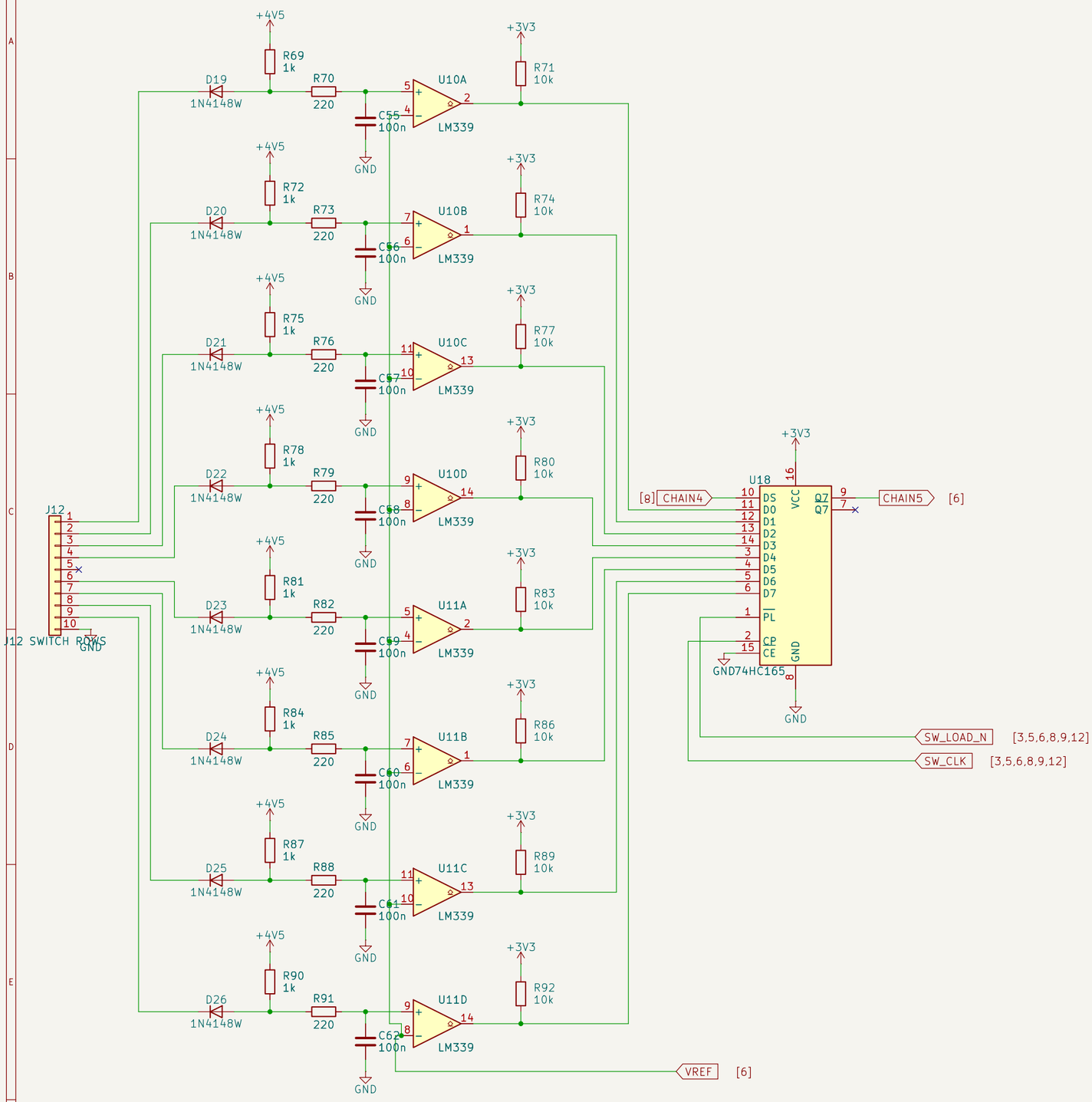
Stern SAM CPU PPUC (CERN-OHL-S v2)

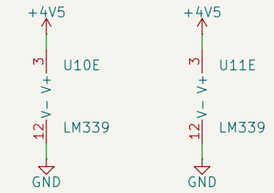
Sheet: /Switch rows 1–8/
File: sw_rows_1.kicad_sch

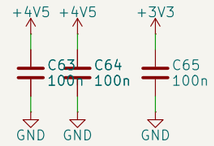
Title: Switch matrix returns 1–8: J6, LM339 front end, 74HC165

Size: A3	Date: 2026-10-06	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 6/13

Per return (ARCHITECTURE.md 5.2–5.3): series diode from the connector, 1 k pull-up to +4.5 V, 220 R + 100 nF, LM339 against VREF (2.25 V). Output 1 = open, 0 = closed, as the ROM expects. Connector pin order follows the original drawing: check it with a meter.







Stern SAM CPU PPUC (CERN-OHL-S v2)

Sheet: /Switch rows 9–16/
File: sw_rows_9.kicad_sch

Title: Switch matrix returns 9–16: J12, LM339 front end, 74HC165

Size: A3 | Date: 2026-10-06 | **Rev: 0.2**
KiCad E.D.A. 10.0.6 | Id: 7/13

Per input (ARCHITECTURE.md 5.2–5.3): 1.5 k pull-up to +4.5 V (switch wetting current) and the original 39 k + 47 pF filter, plus 68 k to ground so the 3.3 V 74HC165 sees 2.8 V (open) or 0 V (closed). Connector pin to D-number order follows the original drawing: check it with a meter.
Memory protect (J2 pin 10) uses the same filter with the original 4.7 k pull-up and goes to input D0 of the 7th 74HC165 (U30, DMD panels sheet).

J2
J2 DEDICATED
J3
J3 DEDICATED
U17
U16
74HC165
+4V5
+3V3
GND
R93 1.5k
R94 39k
R95 68k
C66 47p
R96 1.5k
R97 39k
R98 68k
C67 47p
R99 1.5k
R100 39k
R101 68k
C68 47p
R102 1.5k
R103 39k
R104 68k
C69 47p
R105 1.5k
R106 39k
R107 68k
C70 47p
R108 1.5k
R109 39k
R110 68k
C71 47p
R111 1.5k
R112 39k
R113 68k
C72 47p
R114 1.5k
R115 39k
R116 68k
C73 47p
R117 4.7k
R118 39k
R119 68k
C75 47p
R120 1.5k
R121 39k
R122 68k
C76 47p
R123 1.5k
R124 39k
R125 68k
C77 47p
R126 1.5k
R127 39k
R128 68k
C78 47p
R129 1.5k
R130 39k
R131 68k
C79 47p
R132 1.5k
R133 39k
R134 68k
C80 47p
R135 1.5k
R136 39k
R137 68k
C81 47p
R138 1.5k
R139 39k
R140 68k
C82 47p
R141 1.5k
R142 39k
R143 68k
C83 47p
C74 100n
C84 100n
CHAIN4
[7]
CHAIN2
[6]
MEM_PROTECT
[12]

Stern SAM CPU PPUC (CERN-OHL-S v2)
Sheet: /Dedicated switches 1-16/
File: sw_dedicated_1.kicad_sch
Title: Dedicated switches D1–D16 (J2, J3) and coin door memory protect
Size: A3 | Date: 2026-10-06 | Rev: 0.2
KiCad E.D.A. 10.0.6 | Id: 8/13

Per input (ARCHITECTURE.md 5.2–5.3): 1.5 k pull-up to +4.5 V (switch wetting current) and the original 39 k + 47 pF filter, plus 68 k to ground so the 3.3 V 74HC165 sees 2.8 V (open) or 0 V (closed). Connector pin to D-number order follows the original drawing: check it with a meter. The 8 DIP switches are read through the last 74HC165 of the chain (U14), whose serial input is tied low.

J13 DEDICATED

R144 1.5k
R145 39k
C85 47p
R146 68k
R147 1.5k
R148 39k
C86 47p
R149 68k
R150 1.5k
R151 39k
C87 47p
R152 68k
R153 1.5k
R154 39k
C88 47p
R155 68k
R156 1.5k
R157 39k
C89 47p
R158 68k
R159 1.5k
R160 39k
C90 47p
R161 68k
R162 1.5k
R163 39k
C91 47p
R164 68k
R165 1.5k
R166 39k
C92 47p
R167 68k

U15 74HC165
U14 74HC165

CHAIN2 [8]

SW3 DIP x8

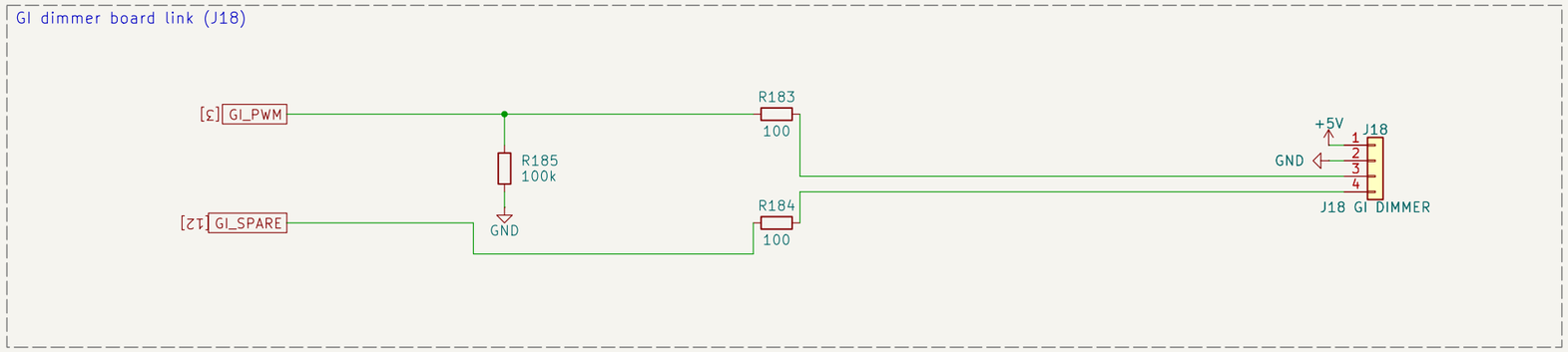
R168 10k
R169 10k
R170 10k
R171 10k
R172 10k
R173 10k
R174 10k
R175 10k

C93 100n
C94 100n

+4V5 +3V3 GND

Stern SAM CPU PPUC (CERN-OHL-S v2)			
Sheet: /Dedicated switches 17–24/			
File: sw_dedicated_2.kicad_sch			
Title: Dedicated switches D17–D24 (J13) and DIP switches			
Size: A3	Date: 2026-10-06		**Rev: 0.2**
KiCad E.D.A. 10.0.6			Id: 9/13

J18 is the default GI dimmer link of ARCHITECTURE.md 10.2: +5 V, GND, GI_PWM, GI_SPARE (GI_SPARE is an on/off output of U31).
The DMD interfaces were removed (2026-10-07): DMD signals do not survive the long cabinet wires.



Stern SAM CPU PPUC (CERN-OHL-S v2)

Sheet: /GI dimmer/
File: display_gi.kicad_sch

Title: J18 GI dimmer header

Size: A3	Date: 2026-10-06	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 10/13

Single +12 V supply (J23). U23 drives left / right in bridge (BTL): no output capacitor, but no speaker wire may go to ground.
J10 keeps the original connector: pins 1 / 2 = left + / right +, pins 3 / 4 (ground on the original) = left − / right −. Check the harness keeps the two speaker returns separate.
J22 = the same two channels as pairs (L+ L− R+ R−). Subwoofer: (L + R) / 2 through a 110 Hz 2nd-order low-pass (U26). U24 drives two bridged outputs on J24 (dual voice coil or two subwoofers).
TDA7297 gain is fixed at 32 dB: 22k / 1.8k input dividers give full output at DAC full scale. AMP_MUTE (RP2354B GPIO37) high mutes all outputs. U23 / U24 need a heatsink.

Stereo amplifier (bridge)

[13] AUD_AMP_L
R187 470
AFL
R188 22k
C104 1u
C103 2.2n
R189 1.8k
U23A TDA7297
AMP_PLAY
AMP_STBY
+12V
U23C
C107 100n
C108 470u
J10 J10 SPEAKERS

[13] AUD_AMP_R
R190 470
AFR
R191 22k
C106 1u
C105 2.2n
R192 1.8k
U23B TDA7297
J22 J22 STEREO

+3V3
FB4 FB
PWR_FLAG
C99 10u

U22 PCM5102A
[13] I2S_LRCK
[13] I2S_DIN
[13] I2S_BCK
AUD_DAC_L [13]
AUD_DAC_R [13]
C96 2.2u
C97 2.2u
C98 2.2u
R186 10k
C100 100n
C101 100n
C102 10u

Subwoofer low-pass

C112 220n
SUB_OUT
C109 1u
R193 20k
R196 10k
C110 1u
R194 20k
C111 100n
U26A OPA1678
U26C C113 100n
R195 100k
R197 10k
R198 10k
C114 10u
U26B OPA1678
C115 1u
R199 22k
C116 1u
R200 1.8k

Standby / mute

+5V
R201 10k AMP_STBY
R202 10k AMP_PLAY
C120 10u
C121 10u
Q4 2N7002
[3] AMP_MUTE
R203 100k

Subwoofer amplifier

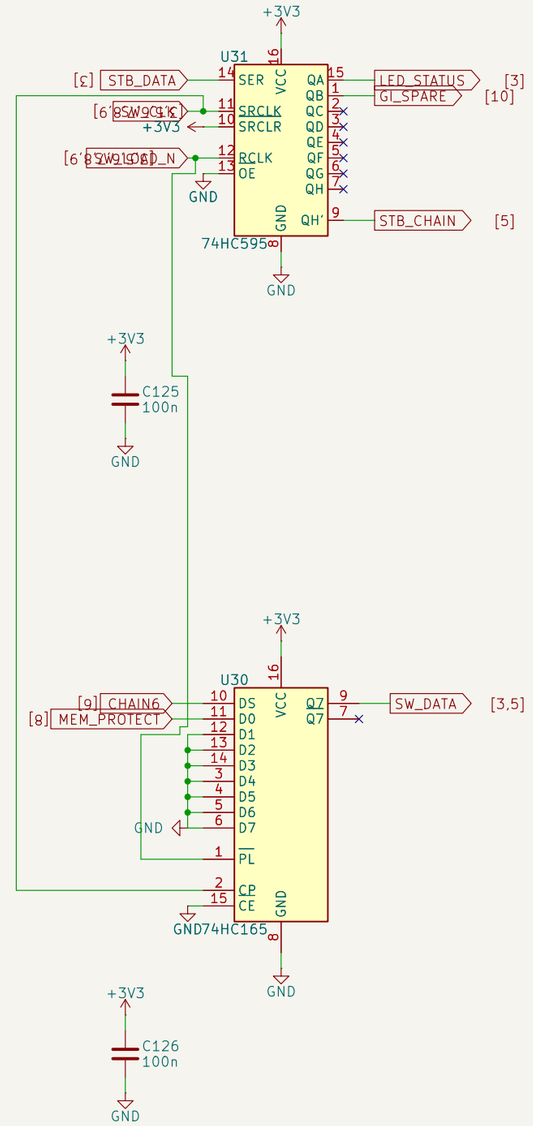
U24A TDA7297
U24C
C117 100n
C118 470u
J24 J24 SUBWOOFER
U24B TDA7297

TP19 DAC_OUTL
TP20 DAC_OUTR
TP21 SUB_OUT
TP22 GND

Stern SAM CPU PPUC (CERN-OHL-S v2)	
Sheet: /Audio/	
File: audio.kicad_sch	
Title: Audio: PCM5102A, TDA7297 stereo + subwoofer (+12 V)	
Size: A3	Date: 2026-10-06
KiCad E.D.A. 10.0.6	Rev: 0.2
	Id: 11/13

U31: first 74HC595 of the strobe chain (status LED, GI_SPARE, 6 spare outputs). U30: last 74HC165 of the switch chain (coin door memory protect + 7 spare inputs). Both share SW_CLK and SW_LOAD_N (165 load / 595 latch on one pulse).
The HD DMD panel outputs that shared this page were removed (2026-10-07).

Switch chain extensions



Stern SAM CPU PPUC (CERN-OHL-S v2)

Sheet: /Switch chain extensions/
File: dmd_panels.kicad_sch

Title: Switch-chain extensions: 7th 74HC165, first 74HC595

Size: A3	Date: 2026-10-06	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 12/13

Half-duplex RS485 on J27 / J29 (in parallel, top-entry RJ45: pin 1 A, pin 2 B, pins 7 / 8 GND_ISO). CA-IS2092VW isolated transceiver: the cable side (A / B, SM712 TVS, 120 R termination on JP2, RJ45 ground) is on GND_ISO, powered by the chip's own isolated DC-DC (VISO 3.3 V, SEL open). Keep GND and GND_ISO copper apart under U32. DE / RE tied: receive by default. Pi on J21: Pi UART3 (GPIO4 TX, GPIO5 RX, GPIO7 RTS = driver enable). Pi remote: RP2354B UART1 (GPIO20 TX, GPIO21 RX, GPIO22 driver enable). Both reach the transceiver through 1 k: only the configured side drives.
J28: I2S from a remote Pi to the PCM5102A (in parallel with J21 pins 12 / 35 / 40); keep that cable short.
J30: line-in (e.g. Pi 4 headphone out). No plug: DAC -> amplifiers through the jack contacts. Plug in: DAC cut off.

Isolated RS485 port (J27 in, J29 out)

+3V3
R212 10k
GND

+3V3 +3V3 +3V3
C127 100n
C128 10u
C129 100n
GND GND GND

C130 100n
C131 10u
C132 100n
PWR_FLAG
GND_ISO GND_ISO GND_ISO

[↓] PI_RS485_RX — R204 1k
[↓] RP_RS485_RX — R205 1k
[↓] PI_RS485_TX — R206 1k
[↓] RP_RS485_TX — R207 1k
[↓] PI_RS485_DE — R208 1k
[↓] RP_RS485_DE — R209 1k

R210 10k
GND

+3V3 U32 CA-IS2092VW
1 VCC
16 VISO
14 VISOIN
6 DI
5 DE
3 RO
4 RE
2 GNDA
11 NC
10 SEL
12 A
13 B
9 GNDB
OSCILLATOR RECTIFIER REGULATOR ENCODE DECODE
GND GND_ISO

R211 120

JP2 TERM

J27 J27 RS485 IN
J29 J29 RS485 OUT
8 7 6 5 4 3 2 1
GND_ISO

D27 SM712
GND_ISO

PWR_FLAG
GND_ISO

I2S input header (J28)

[11] I2S_BCK
[11] I2S_LRCK
[11] I2S_DIN
J28
1 2 3 4 5 6
J28 I2S IN
GND

Analog line-in (J30)

J30
J30 LINE IN
S GND
R AUD_AMP_R [11]
RN AUD_DAC_R [11]
T AUD_AMP_L [11]
TN AUD_DAC_L [11]

Stern SAM CPU PPUC (CERN-OHL-S v2)
Sheet: /RS485 and I2S/
File: rs485_i2s.kicad_sch
Title: Isolated RS485 port (2x RJ45), I2S input header, analog line-in
Size: A3 | Date: 2026-10-06 | **Rev: 0.2**
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